Agents and Skills

Project: Auditable AI Agents
As of: 2026-10-30
This file has two honest parts: what was actually used while this project was prepared, and what has now been prepared for you to reuse. Nothing here is padded.

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1. What was used to prepare the project

1.1 Skills (instruction packs loaded with the Skill tool)

Skill	Used for	How
claude-api	Writing the Claude client (src/app/anthropic_llm.py)	Invoked once. Its guidance shaped the client: model claude-opus-5-5, no forced tool_choice, one tool call at a time, raw model output blocks replayed unchanged.
docx (Anthropic skills)	Building the Anna University Word report	Invoked once. Its notes on page size, tables, numbering and schema validation were followed; its validator checked the file.

1.2 Skill files already in the project (not run)

`.cursor/skills/IEEE-format`, `.cursor/skills/Reasrach-paper-format` (an academic-writer skill) and `.cursor/skills/paradigm-checker` were copied from your first project. They are Cursor rules. I read the two paper-format ones when preparing the report as guidance, but did not run any of the three as a skill. The paradigm checker is written for TypeScript (`const`, no `class`), so it does not apply directly to this Python project; the code follows the same idea (pure functions, immutable data) without being checked by it.

1.3 Subagents

None were started. All work was done directly in the main session. No multi-agent workflow was run.

1.4 Tools used

Tool	Used for
Bash	Running tests, Hardhat, Ollama, the demo, builds
Read, Write, Edit	All code and documents
WebSearch, WebFetch	Finding the Anna University formats and checking every tutorial link
AskUserQuestion	Choosing the local model to download
Terminal tab	One check for an API key (interrupted; the organisation has no key)

Tool	Used for
Task list and wrap-up	Progress tracking

1.5 Agents inside the product (what the system itself runs)

Agent or model	Role
Demo agent (src/domain/agent.py)	A tool-using agent with a calculator, word counter and capital lookup; every step is audited
scripted_llm	Deterministic stand-in model for fast tests
Ollama with qwen3:8b	Real local model, used for the live runs
Claude adapter	Written and unit-tested against a fake client; never run live (no API key)

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2. What has been prepared for reuse

These live in `.claude/` and are picked up by Claude Code when you open the project folder. **They are written but not yet exercised as agents or skills**, so expect to adjust their wording after the first real use.

2.1 Skills (`.claude/skills/`)

Skill	Purpose	Typical request
run-all-tests	Runs both test suites, reports counts and every skip with its reason	"Run all tests"
build-anna-report	Rebuilds the Word report and PDF preview, validates, checks the page count, reminds you to update fields in Word	"Rebuild the report"
verify-citations	Opens each source, checks authors, venue, year and pages, clears the [VERIFY] flags, never fills from memory	"Verify my references"
explain-concept	Teaches one concept: analogy, definition, real code excerpt, a command to try, a self-check question	"Explain Merkle proofs"

2.2 Subagents (`.claude/agents/`)

Agent	Tools	Purpose
citation-verifier	Read, Edit, Bash, WebFetch, WebSearch	Does the work of verify-citations in the background and edits the reference lists
report-reviewer	Read, Grep, Glob, Bash (read-only)	Reviews the generated report against the format guide and checklist; checks every number in Chapters 4 and 5 against tests and runs
tamper-tester	Read, Write, Bash	Red team: invents new attacks against the verifier, writes them as tests, reports any that slip through. Never changes src/

2.3 How to use them

- A skill is invoked by name, for example `/run-all-tests`, or by asking in plain words.
- A subagent is delegated to by name, for example "use the citation-verifier agent on the report references".
- All of them respect the same rules: no invented numbers, no filling citations from memory, and no claims about gas, overhead or scaling until those experiments are run.

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3. Suggested next uses

1. Run `citation-verifier` on `docs/report/phase1-report.md` before the Phase I review.
2. Run `tamper-tester` once; any attack it finds becomes either a fix or a stated limitation in the thesis.
3. Run `report-reviewer` after you fill in your register number, supervisor and dates.